Department of Information Technology
Academic Year 2024 - 2025 (Odd Semester)

Semester, Degree & Branch: V Semester, B.Tech. IT
Course Code & Title: CS3551 & Distributed Computing
Name of the Faculty member (s): Mrs. A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit II / Global State Snapshot Algorithms for FIFO Channels
- **Course Outcome:** The students will be to explore the understand fundamental knowledge of a topic, enable self-learning ability in distributed computing.
- **Activity Chosen:** Think-Pair-Share (TPS)
- **Justification:**
  In unit II, the students need to understand the concepts of Global State Snapshot Algorithms for FIFO Channels. This activity helps the students to understand the Snapshot Algorithms for FIFO Channels of distributed Computing.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
- The topic Global State Snapshot Algorithms for FIFO Channels was given students as shown in figure 1.
- The class students paired with 29 groups. Each student discussed with their pairs for 5 minutes.
- After that, the students were asked to present the topics as shown figure 2.
- Ms. S Swathi team explained the concepts of Snapshot Algorithms for FIFO Channels clearly.
- Next, I called Ms.S. Sobana and their pair, they explained what they study in their book.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO10	P012	PSO1	PSO2	PSO3
CO2	3	2	1	1	2	2	2	3	3	2

(1-Low 2-Mederate 3-High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO8**	**PO10**	**P012**	**PSO1**	**PSO2**	**PSO3**
	3	**2**	**1**	**1**	**2**	**2**	**2**	**3**	**3**	**2**
Justification for correlation	Students could get the knowledge in global state and snapshot recording algorithm using basic concepts of IT.	Students will be able to identity the basic knowledge of the global state and snapshot recording concepts, formulate and analyze the best solution for IT problems,	Students could provide the solution for complex problem with the knowledge of global state and snapshot recording.	Students can elaborate the knowledge of global state mechanism and snapshot recording could be analyzed in distributed computing.	Students could apply to do global state mechanism and snapshot recording by following the ethical principal or rules.	Students to produce the message ordering and snapshot in distributed computing	Students could use the snapshot algorithms, global state and casual ordering mechanism to develop the software.	The basic understanding of the designing will enable the students to develop software to solve the complex computation tasks.	Students could apply the knowledge of features supported by various UI design in developing reliable IT solutions.	Students can be able to apply appreciable knowledge of design and development practices to deliver a quality outcome in software project.

- **Images / Screenshot of the practice:**

Figure 1. Discussion on the topic Global State snapshot Recording

Screen shots

Figure 2. Presentation of students

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

- The students are actively participated in activity
- The said this activity helped to think and recall information due to our communication skills improved.

- ***Benefit of the practice:***
    - The students understand well about Snapshot algorithm recording in distributed computing.
    - Interaction with the faculty member and their class mates during the class was effective.
    - The fear of students was reduced compare when discussing with their friends.
    - It helps the students to think individually about a topic or answer to a given question.
    - With the help of this activity, each student can get clear idea about group communication in distributed computing.
- ***Challenges faced in implementation:***
    - Some students are hesitated to come to dias.
    - The activity was planned for 10 minutes, but it takes 15 minutes because the students hesitate to come dias.

References:

- https://www.geeksforgeeks.org/chandy-lamports-global-state-recording-algorithm/
- https://people.cs.umass.edu/~arun/590CC/lectures/Snapshots.pdf
- https://www.youtube.com/watch?v=CegPJytwsqE
- https://www.youtube.com/watch?v=moSWS1Gha2g

Signature of Faculty Member **HOD**